Ethiopia's Macroeconomic Dilemma:

Exploring the Interplay Between Inflation and Exchange Rates

Anja Jones Luke McDougall Zachary Payne-Thompson Chris Johnson
Group J – Macroeconomics (ECOM181) group project

RQ: Exploring the relationship between inflation and exchange rate devaluation in Ethiopia to fix exchange rate issues and stabilise the macroeconomic environment.

Non-Technical Executive summary

Ethiopia is facing various macroeconomic challenges, with one of the highest inflation rates globally. Inflation has been fueled through global and domestic shocks, high import costs, and supply constraints due to the ongoing political conflict, leaving devastating impacts on the country. The macroeconomic instability risks reversing much of the development progress Ethiopia has made in the past decades. Under current institutional frameworks, the National Bank of Ethiopia (NBE) is faced with the dual challenge of managing high inflation, while maintaining stable exchange rates. In this project we will investigate the trends and drivers of persistent inflation, focusing on exchange rate pressures. This analysis is in light of the IMF suggesting a devaluation in the exchange rate could provide an alternative route in pursuing macroeconomic stability, and reducing the divergence between the official and parallel exchange rate. The NBE aims to use the exchange rate policy as a tool for economic stability, in order to preserve the purchasing power of the national currency, and maintain price stability. However, in light of unprecedented inflation levels, other policies need to be considered above the restrictive exchange rate policy in place. Especially as the overvalued currency has led to shortages in foreign exchange reserves, reduced domestic industrialisation, and caused a growing divergence between the official and parallel exchange rate.

We investigate an expansion of the canonical New Keynesian Model from the closed to the small open economy to investigate inflation dynamics post-exchange rate adjustments. In the context of Ethiopia, supply shocks such as political conflict and global commodity price spikes have exacerbated inflation with an acute impact on import prices. The model, however, implies that optimal monetary policy still remains an inflation targeting regime, rather than one of managing exchange rates which is further supported by literature.

Our empirical analysis, supported by Structural Vector Autoregression (SVAR) modelling, suggests that a devaluation of the Ethiopian Birr might not significantly disrupt trade flows, evidenced by minimal impacts on exports and imports. Furthermore, the anticipated inflationary effects of such a devaluation appear to be neither immediate nor substantial. This finding aligns with the IMF's recommendation, indicating a relatively low risk of inflationary pressures arising from adjustments to the Birr's valuation. Our analysis suggests devaluation would actually provide more stability and is unlikely to cause excessive inflation due to the current misalignment. However, due to the importance of expectations, even small increases in inflation can "run away" when a central bank has low credibility within markets. Thus, we recommend Ethiopia implement three policies; (1) *Devalue the official Ethiopian Birr exchange rate to the parallel rate and float*, (2) *alter the National Bank of Ethiopia's mandate to focus purely on inflation targeting*, and (3) *design and implement an export substituting industrialisation strategy to move the economy from agriculture into industry*. These should restore stability and credibility to usher in steady growth.

1. Introduction

Ethiopia is grappling with a complex set of challenges, including the impact of the COVID-19 pandemic, recurring droughts, locust invasions, internal conflicts, and the repercussions of global events (Government of Ethiopia, 2023, p.17). Although Ethiopia has enjoyed strong economic growth over the past decades and achieved significant advancements in overall development, the rapid growth rate has partly overshadowed underlying macroeconomic vulnerabilities (Ramesh and Abebe, 2016). Currently Ethiopia is crippled by one the highest inflation rates globally, averaging 29% in 2023 (UNDP, 2023). There has been a significant decline in the country's macroeconomic stability, due to inflation, debt, and significant imbalances in the national currency, which are hampering economic growth (UNDP, 2023). Within the complex set of macroeconomic instabilities, Ethiopia faces the dual challenge of managing high inflation, while maintaining a stable inflation rate. The emergence of parallel exchange rates indicates underlying economic issues, caused by foreign exchange shortages, and restrictive exchange rate policies (IMF, 2023). The Government of Ethiopia highlights (2023) how macroeconomic stabilisation is fundamental for recovery and achieved through fixing issues with exchange rates and price distortions, controlling inflation through fiscal and monetary policies and stabilising debt. The IMF echoes these policy priorities and recommends Ethiopia to devalue the currency to fix exchange rate issues and stabilise the macroeconomic environment, which has potential benefits in terms of aligning the official and parallel exchange rates. This paper will explore the relationship between inflation and exchange rate devaluation, in light of the concern that the devaluation of the Ethiopian Birr may further increase the cost of imports, creating a vicious cycle of inflationary pressures.

Background

The risk of exchange rate devaluation further fuelling inflation, could be of specific concern in the context of Ethiopia being heavily import reliant, facing abnormal domestic inflation pressures and supply-side disruptions, as will be outlined in this section. Ethiopia's inflation levels saw a noticeable upward trend in the years after the Covid-19 pandemic, and are nearly double those of the regional averages in Sub-Saharan Africa and East Africa (IMF, 2023., UNDP, 2023). Ethiopia's inflation rate averaged 33.9% in 2022 and 29.1% in 2023, significantly up from 14.4% in pre-pandemic years (2011-2019). The Sub-Saharan African region as a whole is struggling with the crippling effects of double digit inflation with 14.5% and the 15.8% in 2022 and 2023, compared to 6.9% in advanced economies. IMF (2023) stresses how the main policy objective for the region is to control inflation to promote macroeconomic stability.

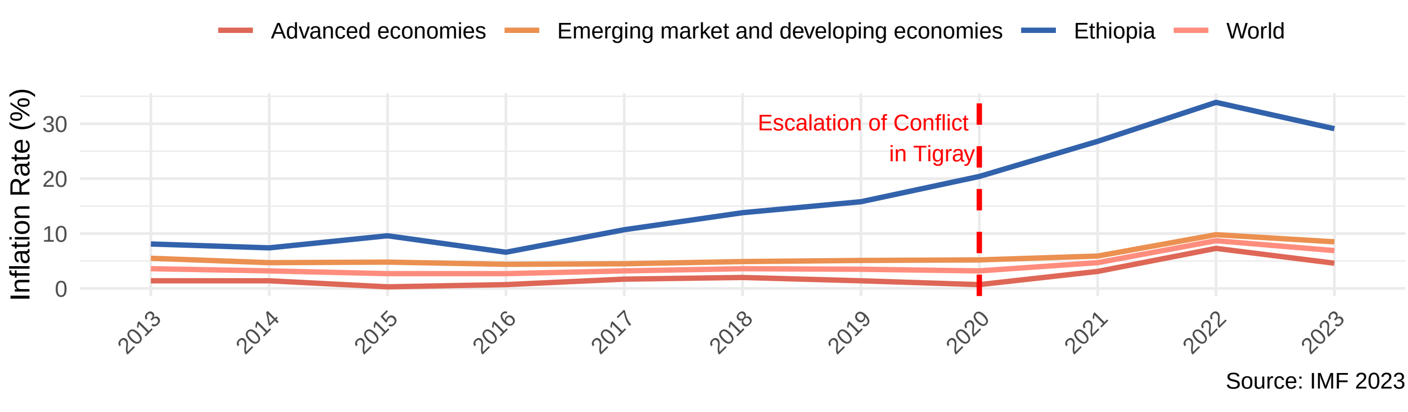

Figure 1: Ethiopia faces one of the highest rates of inflation globally

Inflation in Ethiopia has been caused by repeated domestic and global shocks, fuelling food and non-food components of inflation (UNDP, 2023). Global incidents such as the impact of Covid-19 and Russia's invasion of Ukraine in 2022 have had an adverse impact on developing countries' economic growth and inflation, by increasing global trade costs and price volatility (UNCTAD, 2022., IMF, 2023). The war in Ukraine surged global commodity prices due to Ukraine and Russia's squeeze on exports in critical commodities such as wheat, grains, and oil, increasing food security and poverty in developing countries (Arndt et al.,2023; FSIN 2023). Arndt et al. (2023) demonstrate that the impact of rising global commodity prices on inflation in developing countries is determined by countries' specific production, trade, and consumption patterns. Ethiopia experienced a rise in domestic prices due facing higher import costs, especially prominent due to being characterised as an import reliant economy, as most consumer goods such as food, cereals, manufactured goods, and agriculture inputs are imported (UNDP, 2023). Arndt et al., (2023) and Yalew et al. (2023) finds how the imports of food, fertilisers and fuel especially contributed to recent inflation.

Inflation has been compounded by Ethiopia's own internal political conflicts and weather conditions. The challenging political situation marked by armed conflict and high tensions in certain regions in the country, has caused both a humanitarian and economic crisis and has led to millions of internally displaced people (Government of Ethiopia, 2023). The ongoing internal conflict and associated political disruptions has resulted in abrupt and prolonged supply-side disruptions, leading to a decline in production of goods and services (Tamru et al. 2022). The conflict creates uncertainty around investment and growth, fostering instability in the macroeconomic environment (Regasa et al. 2023, Government of Ethiopia, 2023). This has amplified underlying structural issues and existing supply side constraints related to underproduction, lacking infrastructure and investment (Ndikumana et al., 2022). Structural issues are highlighted by Tamru et al. (2022) in terms of underproduction of food, namely the marketable surplus of grain being undersupplied in light of unmet demand. Ndikumana et al. (2022) find imbalances in the monetary sector, grains sector, and food markets contribute to inflation over the long term and in the short term, structural elements, particularly gaps in cereal production and inflation imported through goods, increase inflation. Both domestic and global factors are contributing to inflation in Ethiopia, as underproduction and a reliance on more expensive imports, fuels it further.

Analysis 1 - Counterfactual modelling on the impact of the conflict on inflation

To understand whether structural issues have materially impacted the Ethiopian economy, we gave conducted counterfactual analysis on inflation. We isolate the impact of the conflict on inflation this report through the pivotal methodology developed by Abadie and Gardeazabal (2003), which utilised the Synthetic Control Method (SCM) to evaluate the economic impact of terrorism in the Basque Country, this report employs a similar approach to understand the inflationary consequences of the conflict escalation in June 2021.

The SCM facilitates the construction of a 'synthetic Ethiopia' to simulate the counterfactual scenario where the conflict did not occur. This SCM analysis relies solely on CPI data, without the use of additional predictors. The synthetic control comprises Haiti (48%), Saint Kitts and Nevis (12%), Zimbabwe (11%), Sierra Leone (10%), and other countries, where the country weights are algorithmically determined to mirror the CPI conditions of Ethiopia prior to the conflict.

Following the escalation of the conflict (indicated by the vertical black line) you can observe a divergence between the consumer price index (CPI) in Ethiopia and those predicted by the synthetic control. In September 2021, CPI in the synthetic control was 25 points below the actual Ethiopia, suggesting a significant inflationary effect attributable to the conflict. However, by the end of 2022 the synthetic control converges back to the observed data, indicating the impact on CPI was temporary. This convergence coincides with the ceasefire to end the internal conflict which was agreed on 2nd Nov 2022.

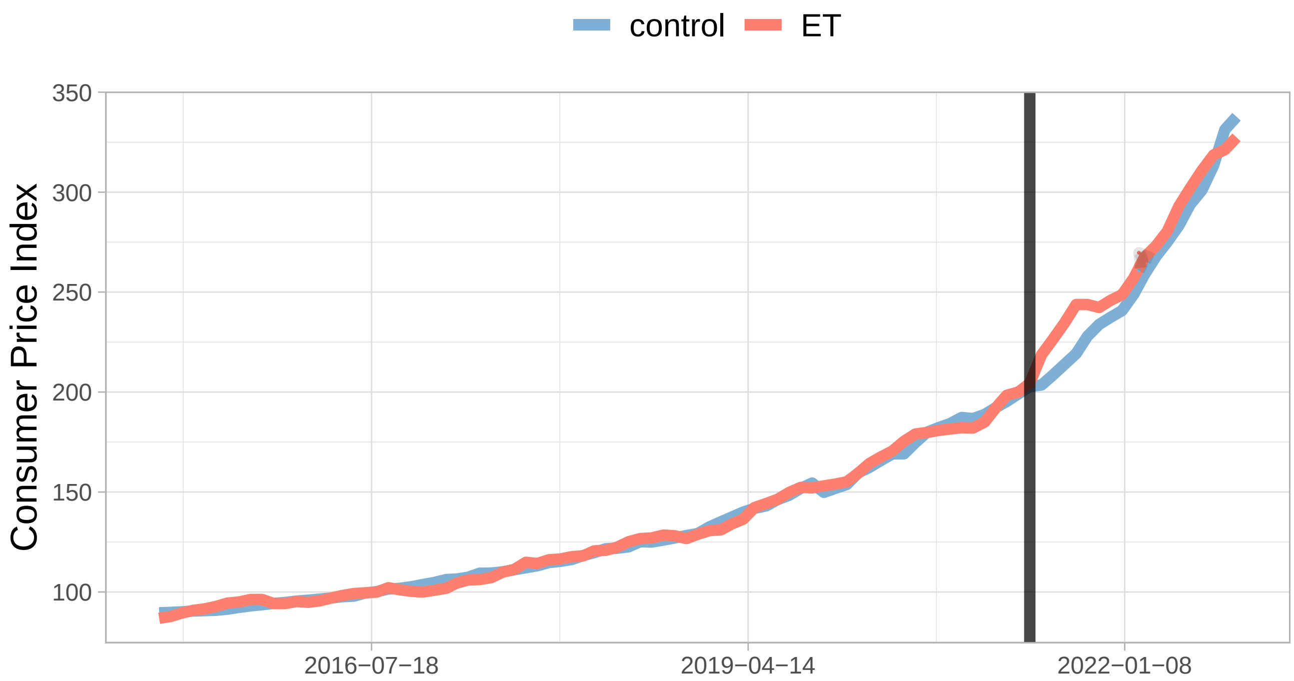

Figure 2: Synthetic control method for the impact of conflict on inflation in Ethiopia

Inflation and exchange rates

Exchange rate pressures contribute to instability, and play a significant role in Ethiopia's current macroeconomic picture, making inflation management harder (IMF, 2023). The Ethiopian currency, the Birr, has depreciated against the dollar steadily since 2016 (IMF, 2023, UNDP, 2023). Notably, huge foreign exchange shortages are leading to a divergence in official and parallel market exchange rate, where the premium in the parallel market is over 80% and close to 100% for some essential items (UNDP, 2023). The shortages are also in part due to the structure and value of exports, reduced earnings for primary commodities in exports, and diminished inflows of foreign direct investment (FDI) (Regasa et al., 2023). Metekia (2023) reports that extensive government spending due to the internal conflicts and political instability, is exacerbating the depletion of foreign exchange reserves alongside the conflict hindering the inflow of foreign currency by restricting tourism and FDI. The diverging gap between the rates, driven by shortages of foreign exchange, maintain the gap and result in inflationary pressures in the economy (Regasa et al., 2023).

The government has attempted to maintain the value of the currency, by operating a managed floating exchange rate regime to maintain import stability, which puts further strain on the foreign exchange reserves (Haile, 2019). The National Bank of Ethiopia (NBE) aims to use the exchange rate policy as a tool for economic stability, in order to preserve the purchasing power of the national currency, and maintain price stability (DFID, 2018). The exchange rate stabilisation policies, including using the exchange rate as a nominal anchor, plays a critical role in curbing inflation and ensuring the economy operates within normal inflationary rates (DFID, 2018). The IMF suggests a devaluation of the currency, which has potential benefits in terms of aligning the official and parallel exchange rates. Moreover, the devaluation of the Ethiopian Birr is concerning policy makers whether it would further increase in the cost of imports, and thus create a vicious cycle of inflationary pressures, especially in light of Ethiopia being an import-reliant economy, facing domestic inflation pressures, and supply-side disruptions.

2. Theory

The New Keynesian Model

To understand optimal monetary policy choices in a small *closed* economy the canonical formulation of the New Keynesian Model with staggered price setting (Calvo 1983) could be used to understand the dynamics of inflation and output.

New Keynesian IS Curve: $x_t = E_t[x_{t+1}] - \phi[i_t - E_t[\pi_{t+1}]]$

New Keynesian Phillips Curve: $\pi_t = \beta E_t[\pi_{t+1}] + kx_t$

In this framework, a Taylor rule could guide interest rate adjustments to directly respond to inflation rates.

Forward-looking interest-rate rule (Taylor rule): $i_t = \delta\pi_t + \varepsilon_t$

In the context of controlling *high* inflation, the Taylor Rule provides a systematic approach for central banks. If inflation rises above the target level, the Taylor Rule formula indicates that the central bank should increase interest rates.

However, this standard formulation is of a *closed* economy and does not explicitly incorporate the role of exchange rates. This omission becomes particularly salient when considering countries like Ethiopia, where monetary policy decisions are closely tied to exchange rate dynamics, especially due to the aforementioned significant impact of import prices on inflation.

To address this gap, we turn to the adaptation of the NK model by Gali and Monacelli (2005). In their work, they extended the basic NK framework to an *open* economy setting by integrating exchange rates. This re-specification offers critical insights into the interplay between exchange rates and monetary policy.

Dynamic IS equation for the open economy:

$$x_t = E_t[x_{t+1}] - \frac{1}{\alpha}(r_t - E_t[\pi_{H,t+1}] - r\bar{r}_t) \tag{1}$$

Where: $r\bar{r}_t \equiv \rho - \sigma_\alpha \Gamma(1-\rho_\alpha)a_t + \alpha\sigma_\alpha(\Theta + \Psi)E_t[\Delta y^*_{t+1}]$ or the Natural Rate of Interest

New Keynesian Phillips Curve for the open economy:

$$\pi_{H,t} + \beta E_t[\pi_{H,t+1}] + k_\alpha x_t \tag{2}$$

Where: $k_\alpha \equiv \lambda(\sigma_\alpha + \psi)$

Here the important differences rely on the inclusion of α which is parameter which describes the degree of openness in an economy. When $\alpha = 0$, the slope of the NKPC becomes the same as in the standard closed economy. And more generally, the equilibrium conditions for the *open* economy can be characterised much in the same way as for the *closed* economy.

Using this framework, we can compare two choices of optimal monetary policy either side of Ethiopia's managed float exchange rate regime.

1) An exchange rate peg: $e_t = 0$

2) Domestic inflation-based Taylor Rule: $r_t = \rho + \phi_\pi \pi_{H,t}$

Via a set of Impulse Response Functions estimated using this theoretical framework, the authors find a trade off between control of the Nominal exchange rate on one hand, and the control of domestic inflation and the output gap on the other. Therefore, even in this model of small *open* economy, optimal monetary policy rules focus on domestic price stability with fully flexible exchange rates. This theoretical framework demonstrates the concept of the "Divine Coincidence" as outlined by Blanchard and Gali (2007). Within the New Keynesian framework, stabilising inflation is analogous to stabilising the output gap. As such, there is no rationale for an exchange rate objective and thus no need to intervene in the foreign exchange market.

Ethiopia's inflation dynamics are significantly influenced by import prices, making the exchange rate a key variable in monetary policy considerations. However, the modified NK model suggests that an inflation-targeting approach remains the optimal strategy. This strategy implies that the central bank should primarily focus on maintaining stable inflation rates, even in the face of exchange rate fluctuations. By doing so, the central bank indirectly contributes to stabilizing the output gap, adhering to the principle of the 'divine coincidence'.

There are, however, important considerations when using this framework. In either case of the closed or open economy, the framework is absent of non-trivial real imperfections. Further research has shown that the introduction of firm-level or labour market dynamics relax some of the strict assumptions in existent in the classic New Keynesian model and result in *some* trade off between inflation and output gaps.

As such, we review the empirical literature to understand how transitioning from a managed exchange rate regime via an exchange rate devaluation could affect a country like Ethiopia.

3. Literature Review

Ethiopia's position can be characterised as a currency crisis. A currency crisis as described in "first generation" models occurs when inconsistencies between exchange rate commitment and domestic economic fundamentals arise (Vlaar, 2000). In Ethiopia's case, on the domestic side, government spending from conflicts and political instability and the lack of foreign reserves when coupled with an unadjusted exchange rate maintained by the National Bank of Ethiopia, has developed inconsistencies causing misalignment between the official and parallel exchange rate.

There is debate on whether returning exchange rates to equilibrium actually causes inflationary pressures. Borensztein & De Gregorio (1999) highlight a key limitation of classical theory is the assumption of flexible prices which often could be invalidated due to factors such as competitive pressures and import substitution. Therefore, they conduct a comprehensive study of 41 currency crises to understand the impact of devaluation on an economy's price level using reduced form regressions that control for previous inflation, the output gap and misalignment of currencies. This study finds that the main determinant of pass-through is the higher the inflation before devaluation. This signifies a potential issue for Ethiopia, as with rampant inflation, an even greater increase could worsen their situation. However, it is also worth noting that an economy operating below potential and/or maintaining an overvalued currency such as Ethiopia is likely to have dampened inflationary pressures. Nonetheless, impacts are time variant, the paper uncovers that initial inflation was low which they pin down to sticky prices but rises after 2-years as businesses are able to enact their flex. These findings are enhanced by Burstein, Eichenbaum, & Rebelo (2002) who look at European, South American and Asian economies during currency crises to investigate the key reasons behind this. Contributing to the sticky prices literature, the authors decompose CPI into tradable and non-tradable goods[1] which they argue and evidence is a key determinant of large devaluation pass-through. By distinguishing between the two types they find that typically non-tradables are slow to adjust prices - limiting inflation - while the change in the prices of tradables falls when looking at import and export prices. They theorise that the price of non-tradables is most important in CPI which is likely welcome

[1]Goods bought and consumed where they are produced

for Ethiopia, however, their import reliance may shift this (Haile, F., 2019). Furthermore, the authors evidence a “flight from quality”. Because imports become more expensive consumers substitute from foreign to domestic goods which not only reduces the proportion of tradables - which are impacted by exchange rates - but also are likely to be lower quality. Because the CPI aims to include quality within its inflation rates this further dampens inflationary pressures. Additionally, for a country like Ethiopia that is trying to industrialise[2], the devaluation may not only lower CPI but also hasten their industrialisation by improving their export capacity. From this we would therefore expect lower CPI post devaluation but this is subject to the substitutability of imports.

The above shows positive signs regarding Ethiopia’s inflation post devaluation but there are a myriad of limitations to applicability such as the focus on mainly Asian, European and South America countries (note, Ivory Coast and Cameroon are included in Borensztein and De Gregorio (1999) but pale in comparison to the other 39 countries). These countries are mainly counted as emerging or developed which means structurally they are comprised of likely different goods sectors and institutions which could impact the applications of previous evidence. Thus, a more targeted assessment of Exchange Rate Pass-Through (EPRT) is likely to uncover some more bespoke factors for developing countries. Barhoumi (2006) conducts a non-stationary panel analysis of 24 countries including 10 African countries to model heterogeneity between exchange rate regimes, tariff barriers and inflation regime (low or high inflation rates). Modelling specifically the price of imports, the author comes to the conclusion that countries with fixed exchange rate regimes, lower tariff barriers and higher ex ante inflation experience the highest pass-through. Therefore, under current arrangements of high imports, high inflation and the National Bank of Ethiopia’s interventions in foreign exchange markets, it is likely that they would experience relatively higher inflation.
Finally, conducting a more micro-level analysis of African countries’ EPRT, Balcilar, Usman, & Agbede (2019) specifically look at Nigeria and South Africa using an autoregressive distributed lag model to understand more nuanced drivers. Large differences are uncovered including oil prices negatively (positively) impacting Nigeria’s (South Africa’s) inflation and output growth positively impacting Nigeria rather than the insignificant effect on South Africa. Due to South Africa being an oil importing nation - similar to Ethiopia - this does suggest potential pressures. However, most importantly, the authors find EPRT was complete for Nigeria while incomplete for South Africa. This raises an important aspect that is typically difficult to model, inflation targeting.

Across all papers a key, sometimes implicit, denominator has been the inflation regime. Within Borensztein & De Gregorio (1999) this was shown in the differences between long and short terms and the outlying effect in the Asian Crises which they pin down to lower inflation expectations and some price controls; in Burstein, Eichenbaum, & Rebelo (2002) the inflation rate post devaluation was much lower in European economies - which were countries with more active inflation control. When combined with Barhoumi (2006) and Balcilar, Usman, & Agbede (2019) it is highly suggestive that when targeting inflation, central banks are more credible which pulls down the later rate of inflation.

Literature has uncovered a complex interaction and it could be seen that devaluation may be a source of inflation given the current institutional environment but this depends on the substitutability of home and foreign goods. However, realigning parallel exchange rates does contain differences which to an extent limits some articles’ findings. According to the IMF (2021), because most of the economy is already operating at the parallel rate it would be unlikely to cause initially large changes in the exchange rate; but the key determinant of long term inflation is the choice of monetary and fiscal policy. This supports the literature review that found monetary policy credibility is a key factor and is likely hampered by the National Bank of Ethiopia’s (NBE) dual, sometimes conflicting, mandate of exchange rate and price level stability. While the NBE uses this to suppress inflation by limiting the pass-through of an open economy, this neglects domestic channels which creates inefficiencies elsewhere. And therefore, when inflation is not hawkishly tackled, agents don’t expect price level changes to slow, raising expectations, causing a higher future rate of inflation and worsening the NBE’s credibility further.

[2]Growth and Transformation Plan II | United Nations in Ethiopia

4. Empirical Estimation

Analysis 2 - Quantitative investigation on exchange rate devaluation on on inflation through SVAR modelling

Ethiopia is negotiating with the IMF to secure a potential loan to alleviate the nation's debt crisis. A key IMF condition for this financial support is the devaluation of the Ethiopian Birr. The IMF suggests this measure to bridge the substantial gap between the official rate (55.35 Birr per USD) and the parallel market rate (approximately 100 Birr per USD), aiming to correct currency over valuation and stimulate the economy. Devaluation can potentially boost export competitiveness by making domestic goods cheaper for foreign buyers. However, it raises concerns in Ethiopia due to its reliance on imports, aligning with the exchange-rate pass-through theory. A weaker domestic currency could increase the local currency cost of imports, potentially leading to higher inflation, especially in import-dependent economies like Ethiopia. Amidst high inflation rates, Ethiopian policy makers are wary of the potential for further inflationary spikes stemming from the IMF-suggested currency devaluation.

Given the lack of direct empirical literature on Ethiopia's exchange rates we have chosen to further empirically assess the impact of a devalued Birr on inflation. Thus, this paper will employ a Structural Vector Autoregression (SVAR) model, capturing the dynamic relationship between the official exchange rate fluctuations, trade flows and inflation rates. The model's strength lies in its ability to capture the direct and indirect consequences of potential policy shifts, offering a comprehensive quantitative evaluation of the inflationary outcomes of the IMF's proposed devaluation policy.

In recent decades SVAR models have emerged as an important research tool for the empirical analysis in macroeconomics. This was largely driven by the work of Sims (1980) who criticised the traditional macro-econometric modelling which were prevalent in the 1970s (Kilian & Lütkepohl, 2017). The SVAR methodology allows for the inclusion of contemporaneous variables and an investigation into the impact of macroeconomic policies and structural shocks.

To estimate a SVAR model we need to impose some restrictions to the parameters in the model to allow for identifications. These restrictions, often based on economic theory or a priori knowledge, enable us to draw causal inferences from the analysis. The variable ordering within our SVAR model - exchange rates, exports, imports, and inflation rate - is grounded in economic theory. Exchange rates are placed first due to their sensitivity to market dynamics and ability to swiftly impact trade balances. Following this, exports and imports are ranked based on their intermediate role in transmitting external shocks to domestic markets. Lastly, inflation is positioned as a lagging variable, reflecting its nature as a gradual response to changes in trade and currency valuations, consistent with exchange rate pass-through theory. This sequence is designed to accurately capture the propagation of a structural shock to exchange rates.

Let Y_t be a vector of endogenous variables at time t comprising Exchange Rates, Exports, Imports and Inflation Rate (where the former three are differenced). The structural form for the SVAR is given by:

$$AY_t = A_1 Y_{t-1} + ... + A_p Y_{t-p} B\epsilon_t \tag{3}$$

Where A is the matrix of contemporaneous coefficients. $A_1, ..., A_p$ are the coefficient matrices for lags 1 to p. B is the matrix linking structural shocks ϵ_t to innovations in the SVAR model. Finally, ϵ_t is a vector of structural shocks. We can also express the SVAR equation in matrix notation as followed:

$$\mathbf{A}\begin{bmatrix}\Delta\text{Exchange Rate}_t\\ \Delta\text{Exports}_t\\ \Delta\text{Imports}_t\\ \text{Inflation Rate}_t\end{bmatrix}=\mathbf{A}_1\begin{bmatrix}\Delta\text{Exchange Rate}_{t-1}\\ \Delta\text{Exports}_{t-1}\\ \Delta\text{Imports}_{t-1}\\ \text{Inflation Rate}_{t-1}\end{bmatrix}+\cdots+\mathbf{A}_p\begin{bmatrix}\Delta\text{Exchange Rate}_{t-p}\\ \Delta\text{Exports}_{t-p}\\ \Delta\text{Imports}_{t-p}\\ \text{Inflation Rate}_{t-p}\end{bmatrix}+\mathbf{B}\epsilon_t \tag{4}$$

The A matrix is structured to reflect a Cholesky decomposition, a lower triangular matrix allowing for the recursive identification of shocks. This structure assumes that a variable can be contemporaneously affected by its own shock and by the shocks of the variables ordered above it. The B matrix, containing zeros off the diagonal, enforces the orthogonality of shocks, meaning that each shock in the system is independent of others at the time it occurs. By specifying how variables interact contemporaneously (e.g., through A matrix and B matrix), we can discern the immediate impact of shocks and trace their subsequent effects through the system.

We estimate the SVAR model with OLS, generate the 95% confidence intervals with bootstrap and produce the associated impulse response functions (IRFs). The latter represent the reaction over time of the variables of interest (in our case, trade flows and inflation rates) to a structural shock (in our case, a devaluation in the Birr). Following Ivanov & Kilian's (2005) paper, we favour the Akaike Information Criterion (AIC) to select the optimal number of lags p as this performs best for monthly data series.

Augmented Dickey-Fuller tests, used to check time series stationarity, reveal that in our case, only inflation rates are stationary in levels. This is critical for SVAR modelling, since it requires stationary data. Consequently, to align with SVAR requirements, we need to difference exchange rates, exports, and imports to achieve stationarity.

Results[3]

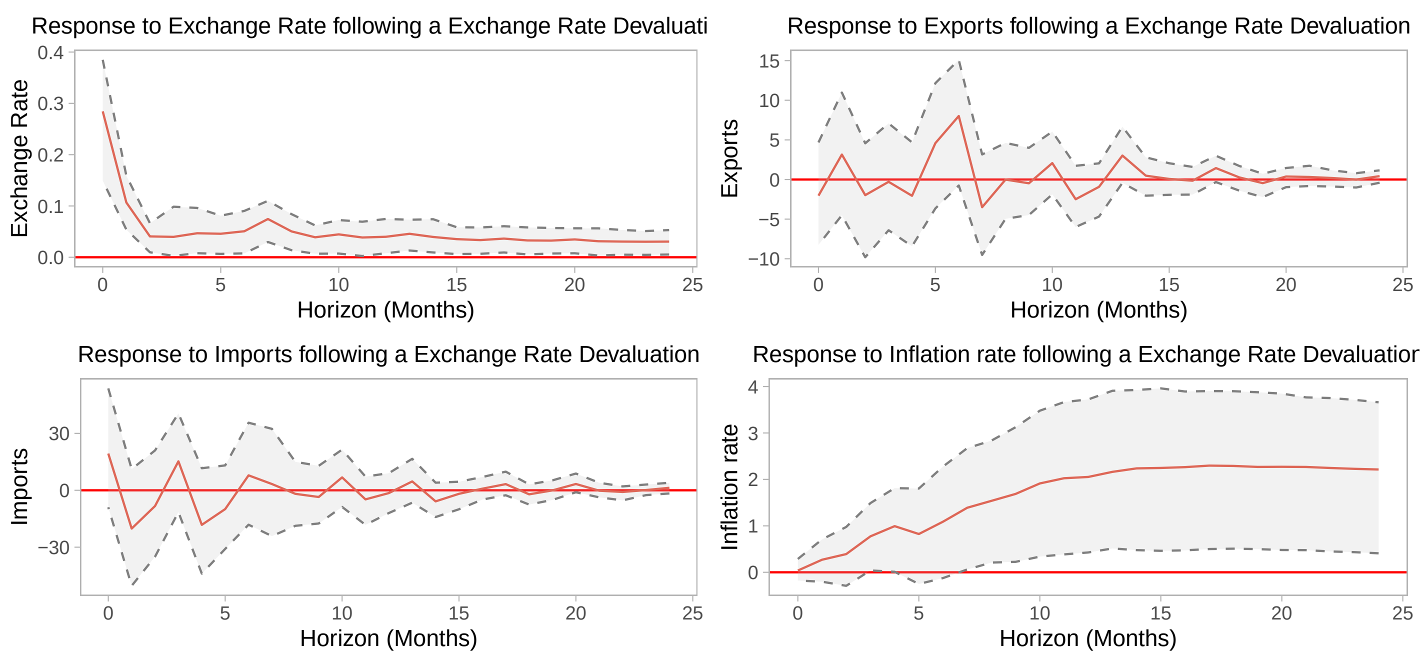

Figure 3: Impulse Response Functions following an Exchange Rate Devaluation

[3]See Annex A for the coefficient estimates of the SVAR and Annex B for the technical derivation.

Figure 3 presents the IRFs from the SVAR model, tracing the variables' responses to a standard deviation shock in exchange rates. The top left panel shows the extent of the persistence in exchange rates that follows the shock. i.e. how fast exchange rates return to the steady state following a devaluation (note the measurement is Birr per USD hence an increase in this measurement is a devaluation of the Birr). The response of exports and imports, shown in the top right and bottom left panel, does not exhibit a statistically significant adjustment, suggesting the absence of empirical support for the exchange rate pass-through effect in this instance. As for inflation rates, a lagged positive response emerges after about 10 months; however, the confidence intervals suggest this effect is only marginally significant at the 95% level, plus the magnitude of the inflationary impact from the devaluation is quite small.

Policy Implications:

The IRFs suggest that a devaluation of the Birr may not alter trade flows, as the response of exports and imports is statistically insignificant. Moreover, the potential inflationary pressures, while present, are not immediate nor substantial in scale. Therefore, this empirical evidence suggests Ethiopian policy makers the risks of igniting inflationary pressures via a devaluation is low. Thus, we recommend Ethiopian officials proceed with the IMF negotiations to secure the loan, even with condition to devalue the Birr.

Limitations:

The SVAR and IRF analysis, while insightful, does have limitations. The differencing of variables such as exchange rates, exports, and imports to achieve stationarity can lead to the loss of long-term information, which may understate the long-run relationships and equilibrium conditions of these economic factors. Additionally, the identification of shocks is based on theoretical assumptions that may not fully capture the complexities of the actual Ethiopian economy. Finally, the wide confidence intervals in the IRFs indicate uncertainty in the estimates, which should be considered when interpreting the results.

5. Policy Recommendations and Conclusion

This paper has shown Ethiopia is in a difficult position. The ongoing warring, lack of foreign exchange reserves and exposure to world crises have culminated in an economy operating in disequilibrium. And for a country that is reliant on trade, uncertainty creates greater risks for producers, consumers and governments which likely raises costs, worsens the inflation levels and limits growth. While not a sweeping resolution, realignment of the parallel and official exchange rate is an important trigger in restoring stability.

When inflation is such a prevalent issue it is key to consider the potential inflationary impacts that result from devaluation. Studying a wide literature on currency crises, empirics suggest a complex set of interactions, but based on IMF research, the currently high parallel and official exchange rate misalignment is likely to dampen any increases in inflation associated with a currency devaluation. However, it is important to note structural differences; focusing on developing and African countries shows that Ethiopia's macroeconomic policy choices such as interventionist exchange rate management and industrial struggles could inadvertently worsen the crisis.

Most importantly, literature has uncovered the dynamics between time periods and evidence has shown the importance of inflation targeting in lowering price levels post devaluation. When inflation is significantly impacted by expectations of future inflation, even small increases can endogenously produce much larger results; and as seen through our bespoke structural VAR modelling, inflationary pressures in the short-term, while low, are larger in the long term and by not wrestling such rises could invalidate our conclusions from devaluation.

We therefore believe a trio of policies should be implemented to not only resolve currency misalignment but also ensure future stability of Ethiopia.

1. Devalue the Ethiopian Birr to a level that corrects the current misalignment and set it to float.
2. Alter the mandate of the National Bank of Ethiopia from their focus on foreign exchange management to inflation targeting.
3. Reform Ethiopia's Growth and Transformation plan to speed up industrialisation that boosts the supply side and creates export substituting growth.

Devaluation is necessary to reduce business uncertainty, and from our analysis, we wouldn't expect disproportionately large inflationary pressures. However, the lack of monetary policy credibility in Ethiopia is likely to be a source of potential longer term inflation and thus the National Bank Of Ethiopia needs to ensure inflation targeting is their primary mandate in order to mitigate the risk of runaway inflation from devaluation. And finally, Ethiopia is currently highly dependent on imports and exports of agriculture and industrial performance has been poor - which our analysis suggests is partly due to foreign exchange management. Reliance on agriculture creates high susceptibility to exchange rate fluctuations which prevents stable growth. A move from agricultural sectors to industry will thus reduce the volatility that comes from commodity exports and allow for less reliance on imports for improving living standards. If these changes are adopted, Ethiopia could resolve persistent instability and allow for continuing their remarkable growth and rise out of poverty.

However, it is important to frame these recommendations in the context of the turbulent political economy of the Horn of Africa. Ethiopia is the highest populated landlocked country in the world; tensions around sea access as a result of political instability exacerbate the impact of global commodity shocks and a general slowdown in the global economy. Furthermore, other domestic crises such as the acute lack of fiscal space and a difficult debt situation limit the levers Ethiopia can pull to mitigate the impact of escalations in conflict. Monetary policy tools can only go so far in securing macroeconomic stability in the context of proliferating internal conflicts, and therefore continued diplomatic efforts to minimise conflicts are a key requirement to mitigate the impact of inflation. Nonetheless, if these challenges are wrestled with, Ethiopia will be well-positioned for socioeconomic development.

References

Abadie, A., & Gardeazabal, J. (2003). The economic costs of conflict: A case study of the Basque country. *American Economic Review*, 93(1), 113–132.

Arndt, C., Diao, X., Dorosh, P., Pauw, K., & Thurlow, J. (2023). The Ukraine war and rising commodity prices: Implications for developing countries. *Glob Food Sec*, 36, 100680. doi:10.1016/j.gfs.2023.100680

Balcilar, M., Usman, O., & Agbede, E. A. (2019). Revisiting the Exchange Rate Pass-Through to Inflation in Africa's Two Largest Economies: Nigeria and South Africa. *African Development Review*, 33(2), 245-247.

Barhoumi, K. (2006). Differences in long run exchange rate pass-through into import prices in developing countries: An empirical investigation. *Economic Modelling*, 23, 926-951.

Blanchard, O., & Galí, J. (2007). Real Wage Rigidities and the New Keynesian Model. *Journal of Money, Credit and Banking*, 39(Supplement 1), 35-65. Retrieved from https://www.jstor.org/stable/4123055

Borensztein, E., & De Gregorio, J. (1999). Devaluation and Inflation After Currency Crises. *International Monetary Fund Working Paper*, 1-34.

Burstein, A., Eichenbaum, M. S., & Rebelo, S. (2002). Why are Rates of Inflation so low after large devaluations? *National Bureau of Economic Research Working Paper*.

Department for International Development (DFID). *Ethiopia: Foreign Exchange for Businesses. Business Environment Reform Facility (BERF).* https://assets.publishing.service.gov.uk/media/5c90aa9240f0b633fc95f6d4/BERF_Ethiopia_Foreign_Exchange_for_Businesses_.pdf. Accessed 16/02/2024.

Food Security Information Network. (2023). *Global Report on Food Crises 2023: Brief.* Retrieved fromhttps://www.fsinplatform.org/sites/default/files/resources/files/GRFC2023-brief-EN.pdf

Galí, J., & Monacelli, T. (2005). Monetary Policy and Exchange Rate Volatility in a Small Open Economy. *Review of Economic Studies*, 72, 707-734. The Review of Economic Studies Limited.

Gray, S. T. (2021). Recognizing Reality—Unification of Official and Parallel Market Exchange Rates. *International Monetary Fund Working Paper.*

Haile, F. (2019). The Exchange Rate: Why It Matters for Structural Transformation and Growth in Ethiopia. *World Bank Working Paper.* https://documents1.worldbank.org/curated/en/898821559134798352/pdf/The-Exchange-Rate-Why-It-Matters-for-Structural-Transformation-and-Growth-in-Ethiopia.pdf

International Monetary Fund. (2023). *Economic Outlook for Sub-Saharan Africa.*

International Monetary Fund. (2023, October 16). *Regional Economic Outlook for Sub-Saharan Africa, October 2023.* Retrieved fromhttps://www.imf.org/en/Publications/REO/SSA/Issues/2023/10/16/regional-economic-outlook-for-sub-saharan-africa-october-2023

International Monetary Fund. (2023). *Regional Economic Outlook: Managing Exchange Rate Pressures in Sub-Saharan Africa—Adapting to New Realities.* Retrieved fromhttps://www.imf.org/-/media/Files/Publications/REO/AFR/2023/April/English/ExchangeNote.ashx

Ivanov, V., & Kilian, L. (2005). A Practitioner's Guide to Lag Order Selection for VAR Impulse Response Analysis. *Studies in Nonlinear Dynamics & Econometrics*, 9(1).

Kilian, L., & Lütkepohl, H. (2017). Structural Vector Autoregressive Analysis. *Cambridge University Press.*

Metekia, D. (2023). Impact of Government Spending on Foreign Exchange Reserves in Ethiopia. *Economic Analysis and Policy*, 54(1), 112-130.

Ndikumana, L., Nkurunziza, J. D., Sánchez Martín, M. E., Mulugeta, S., & Kelbore, Z. G. (2022). Monetary, fiscal, and structural drivers of inflation in Ethiopia: New empirical evidence from time series analysis. *Review of Development Economics*, 27(2), 924-962.

Ramesh, R., & Abebe, A. (2016). Has Economic Growth Contributed to Human Development in Ethiopia? *Journal of Asian and African Studies*, 51(6).https://doi.org/10.1177/00219096145553

Sims, C. A. (1980a). Macroeconomics and Reality. *Econometrica*, 48, 1–48.

Tamru, S., Lemma, S., Taffesse, A. S., & Abate, G. T. (2022). Inflation in Ethiopia: Supply-side drivers. *International Growth Centre Policy Brief.* Retrieved fromhttps://www.theigc.org/sites/default/files/2022/08/Tamru-et-al-Policy-Brief-August-2022.pdf

United Nations Conference on Trade and Development (UNCTAD). (2022). *The Impact on Trade and Development of the War in Ukraine.* Geneva: UNCTAD. Retrieved fromhttps://unctad.org/system/files/officialdocument/osginf2022d1_en.pdf

Vlaar, P. J. (2000). Currency Crisis Models for Emerging Markets. *Bank for International Settlements Working Paper.*

Yalew, A. W., Nechifor, V., & Ferrari, E. (2023). Implications of Russia's war against Ukraine for African economies: A CGE analysis for Ethiopia (JRC135512). *European Commission Working Paper.*

Annex A

Tables for SVAR Estimation Results:

Table 1: Estimated A Matrix

	exchange_rate	exports	imports	inflation_rate
exchange_rate	1.0000	0.0000	0.0000	0
exports	7.1727	1.0000	0.0000	0
imports	-65.9112	0.3119	1.0000	0
inflation_rate	-0.1636	-0.0000	0.0005	1

The diagonal elements of the A matrix are all set equal to 1, indicating that each variable (exchange rate, exports, imports, inflation rate) is influenced by its own past values. The off-diagonal elements represent the contemporaneous relationships between these variables. Notably, a relatively high negative coefficient for imports with respect to exchange rates (-65.9112) suggests that an increase in the exchange rate (devaluation of the currency) is associated with a substantial decrease in imports. This could be due to higher costs of imported goods when the domestic currency is weaker. The other coefficients in this matrix are relatively smaller, indicating less immediate interdependence between these variables.

Table 2: Estimated B Matrix

	exchange_rate	exports	imports	inflation_rate
exchange_rate	0.2844	0.0000	0.0000	0.0000
exports	0.0000	48.1700	0.0000	0.0000
imports	0.0000	0.0000	186.4000	0.0000
inflation_rate	0.0000	0.0000	0.0000	2.2550

The B matrix shows the magnitude and direction of the structural shocks' impacts on the variables. The exchange rate has a moderate shock value (0.2844), indicating that unexpected changes in the exchange rate can have a noticeable impact. A large shock value for imports (186.4) suggests that unexpected changes in imports can significantly influence the economy, possibly due to Ethiopia's reliance on imported goods. The shock value for inflation (2.255) is relatively moderate, indicating that inflation is somewhat sensitive to unanticipated economic shocks. The zeros in the B matrix indicate that shocks to one variable do not directly affect the others at the same time (reflecting the assumed independence of shocks).

Annex B

Technical Derivation of the Synthetic Control Method

Mathematically, the synthetic control for Ethiopia, denoted as $\text{Ethiopia}_t^{\text{synth}}$, is derived as follows:

$$\text{Ethiopia}_t^{\text{synth}} = \sum_{j=1}^{J} w_j \cdot \text{Control}_{jt}$$

Here, t denotes the time period before the intervention, J is the number of control units, Control_{jt} represents the CPI of control unit j at time t, and w_j are the weights assigned to each control unit. These weights are calculated to minimize the difference in pre-conflict CPI between Ethiopia and the synthetic Ethiopia, typically through a least squares optimization problem:

$$\begin{aligned} & \underset{w}{\arg\min} \\ & \sum_{t=1}^{T_{\text{pre}}} \left(\text{Ethiopia}_t - \sum_{j=1}^{J} w_j \cdot \text{Control}_{jt} \right)^2 \\ \text{subject to} \quad & \sum_{j=1}^{J} w_j = 1 \quad \text{and} \quad w_j \geq 0 \quad \text{for all} \quad j \end{aligned}$$

In this optimization, T_{pre} represents the final pre-conflict period. The objective is to find the set of weights $\{w_j\}$ that minimizes the squared difference between the actual CPI in Ethiopia and the weighted average of the CPI in the control units, thereby constructing a synthetic Ethiopia that closely resembles the actual Ethiopia before the internal conflict.